

Tokenization for Large Language Models

Summer 2026, LMU Munich

Prof. Dr. Valentin Hofmann

Roadmap for Today

Housekeeping and Recap

Character Encoding Basics

Byte-Pair Encoding

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Seminar Website

Tokenization for Large Language Models 2026

Schedule and materials for the seminar

[View My GitHub Profile](#)

This is the website for the seminar [Tokenization for Large Language Models](#), offered at the Center for Information and Language Processing (CIS), LMU Munich, in summer 2026.

Instructor: [Prof. Dr. Valentin Hofmann](#)

Contact: firstname.lastname@lmu.de

Date	Topic	Slides
17.04.2026	Introduction, LLM recap	Link
24.04.2026	Character encoding, BPE	
08.05.2026	Unigram, topics presentation	

<https://tokenization-seminar-2026.github.io/>

Dates for Two-Day Block

- Proposed dates: 20.07.2026, 21.07.2026
- Any objections?

Word-Level and Character-Level Tokenization

	Word-level	Character-level
Pros		
Cons		

Word-Level and Character-Level Tokenization

	Word-level	Character-level
Pros	• Relatively simple to implement • Exact lexical representations	
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Cons	• Large LLM vocabularies are computationally very expensive • Does not work for languages without whitespace • Heap's law: full coverage of all words impossible → many out-of-vocabulary words	

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Cons	• Large LLM vocabularies are computationally very expensive • Does not work for languages without whitespace • Heap's law: full coverage of all words impossible → many out-of-vocabulary words	• LLM needs to learn lexical representations as a function of characters • Difficult to train in practice

Questions?

Roadmap for Today

Housekeeping and Recap ✓

Character Encoding Basics

Byte-Pair Encoding

Characters

- There are **many** characters (e.g., more than 50K characters in Chinese alone)

词法分析 [编辑]

条目 讨论 汉 汉 不转换 阅读 编辑 查看历史 工具

维基百科，自由的百科全书

 此條目已列出參考資料，但文内引註不足，部分內容的來源仍然不明。（2026年3月24日）
请加上合适的文内引註加以改善。

词法分析（英語：**lexical analysis**）是**计算机科学**中将字符序列转换为**词元**（英語：token，台湾作**符元**，也有译为**标记**或**记号**）序列的过程。进行词法分析的程序或者函数叫作**词法分析器**（lexical analyzer，简称lexer），也叫**扫描器**（scanner）。词法分析器一般以函数的形式存在，供**语法分析器**调用。

Characters

- There are **many** characters (e.g., more than 50K characters in Chinese alone)
- **Unicode**:
 - Standardized character set
 - 1,114,112 individual **code points**, out of which 159,801 currently map to a character
 - Different encodings (e.g., UTF-8, UTF-16, UTF-32)

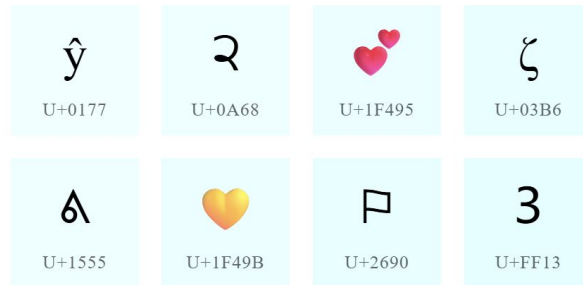
词法分析 [编辑] 🌐 34种语言

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Bits and Bytes

- **Bit**: smallest unit of data representing **single binary value** (0 or 1)

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- **Byte**: unit of data consisting of **eight bits** (e.g., 00100110)
- There are 2^8 (= 256) different bytes

ASCII

- Each character encoded with **seven bits** (i.e., 128 code points)
- Heavy focus on **English**

b7b6b5b4b3b2b1 Bits Column Row					0	1	2	3	4	5	6	7	
	0	1	2	3	4	5	6	7					
0	0	0	0	0	0	NUL	DLE	SP	0	@	P	`	p
0	0	0	0	1	1	SOH	DC1	!	1	A	Q	a	q
0	0	0	1	0	2	STX	DC2	"	2	B	R	b	r
0	0	0	1	1	3	ETX	DC3	#	3	C	S	c	s
0	1	0	0	0	4	EOT	DC4	\$	4	D	T	d	t
0	1	0	0	1	5	ENQ	NAK	%	5	E	U	e	u
0	1	0	1	0	6	ACK	SYN	&	6	F	V	f	v
0	1	0	1	1	7	BEL	ETB	'	7	G	W	g	w
1	0	0	0	0	8	BS	CAN	(	8	H	X	h	x
1	0	0	0	1	9	HT	EM	)	9	I	Y	i	y
1	0	0	1	0	10	LF	SUB	*	:	J	Z	j	z
1	0	0	1	1	11	VT	ESC	+	;	K	[	k	{
1	0	1	0	0	12	FF	FS	,	<	L	\	l	
1	0	1	0	1	13	CR	GS	—	=	M	]	m	}
1	0	1	1	0	14	SO	RS	.	>	N	^	n	~
1	0	1	1	1	15	SI	US	/	?	O	_	o	DEL

UTF-8 Overview

- Variable-length encoding of one to four bytes
- One byte: ASCII characters (backward compatibility)
- Two bytes: almost all Latin and many non-Latin alphabets (e.g., Arabic)
- Three bytes: remaining Basic Multilingual Plane (BMP), including Chinese
- Four bytes: non-BMP code points (e.g., emojis)

UTF-8 Encoding Table

Byte 1	Byte 2	Byte 3	Byte 4
0yyyyzzzz			

ASCII

UTF-8 Encoding Table

Byte 1	Byte 2	Byte 3	Byte 4
0yyyyzzzz			
110xxxxyy	10yyzzzz		

ASCII

(Non-)Latin alphabets

UTF-8 Encoding Table

Byte 1	Byte 2	Byte 3	Byte 4
0yyyyzzzz			
110xxxxyy	10yyzzzz		
1110www	10xxxxyy	10yyzzzz	

ASCII

(Non-)Latin alphabets

BMP (e.g., Chinese)

UTF-8 Encoding Table

Byte 1	Byte 2	Byte 3	Byte 4
0yyyzzzz			
110xxxxyy	10yyzzzz		
1110wwww	10xxxxxyy	10yyzzzz	
11110uvv	10vvwwww	10xxxxxyy	10yyzzzz

ASCII

(Non-)Latin alphabets

BMP (e.g., Chinese)

Non-BMP (e.g., emojis)

Questions?

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Subword-Level Tokenization

- Idea: split the input text into a mixed sequence of words, subwords, and characters (or bytes)

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- Pros:
 - Relatively small token vocabularies (30K to 100K tokens)
 - Exact lexical representations for frequent words
 - Less frequent words represented as a (typically short) sequence of subwords and characters

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- Idea: split the input text into a mixed sequence of words, subwords, and characters (or bytes)
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 - Relatively small token vocabularies (30K to 100K tokens)
 - Exact lexical representations for frequent words
 - Less frequent words represented as a (typically short) sequence of subwords and characters
- Cons:
 - Most complex tokenization approach

Subword-Level Tokenizer Zoo

- Three widely used subword-level tokenizers:
 - **Byte-pair encoding** (BPE; Sennrich et al., 2016): GPT, Llama, Olmo, Qwen
 - **WordPiece** (Wu et al., 2016): BERT, DistilBERT, ELECTRA
 - **Unigram** (Kudo, 2018): ALBERT, XLNet, T5

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- Two key questions:
 - **Vocabulary creation (training)**: how to determine the set of tokens?
 - **Segmentation strategy (inference)**: how to segment texts into tokens?

BPE Vocabulary Creation

- Originally a text compression algorithm (Gage, 1994)
- Basic functioning:
 - Given training corpus, create **base vocabulary consisting of all characters** (= initial set of tokens)
 - Iteratively **merge pair of adjacent tokens with highest frequency** into new token
 - Stop when vocabulary has predefined size

BPE Pseudocode

```
function BYTE-PAIR ENCODING(strings  $C$ , number of merges  $k$ ) returns vocab  $V$   
  
   $V \leftarrow$  all unique characters in  $C$            # initial set of tokens is characters  
  for  $i = 1$  to  $k$  do                             # merge tokens  $k$  times  
     $t_L, t_R \leftarrow$  Most frequent pair of adjacent tokens in  $C$   
     $t_{NEW} \leftarrow t_L + t_R$                    # make new token by concatenating  
     $V \leftarrow V + t_{NEW}$                          # update the vocabulary  
    Replace each occurrence of  $t_L, t_R$  in  $C$  with  $t_{NEW}$    # and update the corpus  
  return  $V$ 
```

Training Data

Proof of the Milky Way consisting of many stars came in 1610 when Galileo Galilei used a telescope to study the Milky Way and discovered that it is composed of a huge number of faint stars.

Given training data D

Training Data

```
{Proof, _of, _the, _Milky,  
_Way, _consisting, _of,  
_many, _stars, _came, _in,  
_1610, _when, _Galileo,  
_Galilei, _used, _a,  
_telescope, _to, _study,  
_the, _Milky, _Way, _and,  
_discovered, _that, _it,  
_is, _composed, _of, _a,  
_huge, _number, _of,  
_faint, _stars.}
```

Pretokenize D by splitting on whitespace

Training Data

_ P r o o f, _ o f, _ t h
e, _ M i l k y, _ W a y, _
c o n s i s t i n g, _ o f,
_ m a n y, _ s t a r s, _ c
a m e, _ i n, _ 1 6 1 0, _
w h e n, _ G a l i l e o, _
G a l i l e i, _ u s e d, _
a, _ t e l e s c o p e, _ t
o, _ s t u d y, _ t h e, _
M i l k y, _ W a y, _ a n
d, _ d i s c o v e r e d, _
t h a t, _ i t, _ i s, _ c
o m p o s e d, _ o f, _ a,
_ h u g e, _ n u m b e r, _
o f, _ f a i n t, _ s t a r
s .

Split D into sequence of bytes

Training Data

_ P r o o f, _ o f, _ t h
e, _ M i l k y, _ W a y, _
c o n s i s t i n g, _ o f,
_ m a n y, _ s t a r s, _ c
a m e, _ i n, _ 1 6 1 0, _
w h e n, _ G a l i l e o, _
G a l i l e i, _ u s e d, _
a, _ t e l e s c o p e, _ t
o, _ s t u d y, _ t h e, _
M i l k y, _ W a y, _ a n
d, _ d i s c o v e r e d, _
t h a t, _ i t, _ i s, _ c
o m p o s e d, _ o f, _ a,
_ h u g e, _ n u m b e r, _
o f, _ f a i n t, _ s t a r
s .

Pair counts

_ t	12335282
t h	10067390
_ a	9319062
h e	8771183
i n	8024060
e r	6517430
a n	6315205
r e	6031043
o n	5261131
_ i	5209828

Vocabulary

Training Data

_ P r o o f, _ o f, _ t h
e, _ M i l k y, _ W a y, _
c o n s i s t i n g, _ o f,
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G a l i l e i, _ u s e d, _
a, _ t e l e s c o p e, _ t
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M i l k y, _ W a y, _ a n
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n s i s t i n g, _ o f, _ m
a n y, _ s t a r s, _ c a m
e, _ i n, _ 1 6 1 0, _ w h
e n, _ G a l i l e o, _ G a
l i l e i, _ u s e d, _ a,
_t e l e s c o p e, _t o, _
s t u d y, _t h e, _ M i l
k y, _ W a y, _ a n d, _ d
i s c o v e r e d, _t h a
t, _ i t, _ i s, _ c o m p
o s e d, _ o f, _ a, _ h u
g e, _ n u m b e r, _ o f,
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l i l e i, _ u s e d, _ a,
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s t u d y, _t h e, _ M i l
k y, _ W a y, _ a n d, _ d
i s c o v e r e d, _t h a
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Vocabulary

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e n, _ G a l i l e o, _ G a
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Vocabulary

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_ s	5035505
_ w	4523998

Vocabulary

_t
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_ M i l k y, _ W a y, _ c o
n s i s t i n g, _ o f, _ m
a n y, _ s t a r s, _ c a m
e, _ i n, _ 1 6 1 0, _ w h
e n, _ G a l i l e o, _ G a
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s t u d y, _t h e, _ M i l
k y, _ W a y, _a n d, _ d i
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_t
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_ P r o o f, _ o f, _t h e,
_ M i l k y, _ W a y, _ c o
n s i s t i n g, _ o f, _ m
a n y, _ s t a r s, _ c a m
e, _ i n, _ 1 6 1 0, _ w h
e n, _ G a l i l e o, _ G a
l i l e i, _ u s e d, _a,
_t e l e s c o p e, _t o, _
s t u d y, _t h e, _ M i l
k y, _ W a y, _a n d, _ d i
s c o v e r e d, _t h a t,
_ i t, _ i s, _ c o m p o s
e d, _ o f, _a, _ h u g e,
_ n u m b e r, _ o f, _ f a
i n t, _ s t a r s .

Pair counts

h e	8771183
i n	8024060
_t h	7897058
e r	6517430
r e	6031043
o n	5261131
_ i	5209828
_ o	5163783
_ s	5035505
_ w	4523998

Vocabulary

_t
_a
he

Training Data

_ P r o o f, _ o f, _t he,
_ M i l k y, _ W a y, _ c o
n s i s t i n g, _ o f, _ m
a n y, _ s t a r s, _ c a m
e, _ i n, _ 1 6 1 0, _ w he
n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _t he, _ M i l k
y, _ W a y, _a n d, _ d i s
c o v e r e d, _t h a t, _
i t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a i
n t, _ s t a r s .

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Training Data

_ P r o o f, _ o f, _t he,
_ M i l k y, _ W a y, _ c o
n s i s t i n g, _ o f, _ m
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n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _t he, _ M i l k
y, _ W a y, _a n d, _ d i s
c o v e r e d, _t h a t, _
i t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
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n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _t he, _ M i l k
y, _ W a y, _a n d, _ d i s
c o v e r e d, _t h a t, _
i t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a i
n t, _ s t a r s .

Pair counts

i n	8024060
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_t he	5605612
e r	5279258
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_ i	5209828
_ o	5163783
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Vocabulary

_t
_a
he

Training Data

_ P r o o f, _ o f, _t he,
_ M i l k y, _ W a y, _ c o
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a n y, _ s t a r s, _ c a m
e, _ i n, _ 1 6 1 0, _ w he
n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _t he, _ M i l k
y, _ W a y, _a n d, _ d i s
c o v e r e d, _t h a t, _
i t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a i
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Vocabulary

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he

Training Data

_ P r o o f, _ o f, _t he,
_ M i l k y, _ W a y, _ c o
n s i s t i n g, _ o f, _ m
a n y, _ s t a r s, _ c a m
e, _ i n, _ 1 6 1 0, _ w he
n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _t he, _ M i l k
y, _ W a y, _a n d, _ d i s
c o v e r e d, _t h a t, _
i t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
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n t, _ s t a r s .

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Training Data

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n s i s t i n g, _ o f, _ m
a n y, _ s t a r s, _ c a m
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n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _t he, _ M i l k
y, _ W a y, _a n d, _ d i s
c o v e r e d, _t h a t, _
i t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a i
n t, _ s t a r s .

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Vocabulary

_t
_a
he
in

Training Data

_ P r o o f, _ o f, _t he,
_ M i l k y, _ W a y, _ c o
n s i s t i n g, _ o f, _ m
a n y, _ s t a r s, _ c a m
e, _ i n, _ 1 6 1 0, _ w he
n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _t he, _ M i l k
y, _ W a y, _a n d, _ d i s
c o v e r e d, _t h a t, _
i t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a
i n t, _ s t a r s .

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Vocabulary

_t
_a
he
in

Training Data

_ P r o o f, _ o f, _t he,
_ M i l k y, _ W a y, _ c o
n s i s t i n g, _ o f, _ m
a n y, _ s t a r s, _ c a m
e, _ i n, _ 1 6 1 0, _ w he
n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _t he, _ M i l k
y, _ W a y, _a n d, _ d i s
c o v e r e d, _t h a t, _
i t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a
i n t, _ s t a r s .

Pair counts

r e	6031043
_t he	5605612
e r	5279258
o n	5261131
_ o	5163783
_ s	5035505
_ w	4523998
a t	4424733
o r	4162447
e s	4010515

Vocabulary

_t
_a
he
in

Training Data

_ P r o o f, _ o f, _t he,
_ M i l k y, _ W a y, _ c o
n s i s t i n g, _ o f, _ m
a n y, _ s t a r s, _ c a m
e, _ i n, _ 1 6 1 0, _ w he
n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _t he, _ M i l k
y, _ W a y, _a n d, _ d i s
c o v e r e d, _t h a t, _
i t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a
i n t, _ s t a r s .

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r e	6031043
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o r	4162447
e s	4010515

Vocabulary

_t
_a
he
in

Training Data

_ P r o o f, _ o f, _t he,
_ M i l k y, _ W a y, _ c o
n s i s t i n g, _ o f, _ m
a n y, _ s t a r s, _ c a m
e, _ i n, _ 1 6 1 0, _ w he
n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _t he, _ M i l k
y, _ W a y, _a n d, _ d i s
c o v e r e d, _t h a t, _
i t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a
i n t, _ s t a r s .

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r e	6031043
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_ o	5163783
_ s	5035505
_ w	4523998
a t	4424733
o r	4162447
e s	4010515

Vocabulary

_t
_a
he
in
re

Training Data

_ P r o o f, _ o f, _t he,
_ M i l k y, _ W a y, _ c o
n s i s t i n g, _ o f, _ m
a n y, _ s t a r s, _ c a m
e, _ i n, _ 1 6 1 0, _ w he
n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _t he, _ M i l k
y, _ W a y, _a n d, _ d i s
c o v e r e d, _t h a t, _ i
t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a
i n t, _ s t a r s .

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Vocabulary

_t
_a
he
in
re

Training Data

_ P r o o f, _ o f, _t he,
_ M i l k y, _ W a y, _ c o
n s i s t i n g, _ o f, _ m
a n y, _ s t a r s, _ c a m
e, _ i n, _ 1 6 1 0, _ w he
n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _t he, _ M i l k
y, _ W a y, _a n d, _ d i s
c o v e r e d, _t h a t, _ i
t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a
i n t, _ s t a r s .

Pair counts

_t he	5605612
o n	5261131
_ o	5163783
_ s	5035505
e r	4754849
_ w	4523998
a t	4424733
o u	3838417
_ c	3831635
n d	3811435

Vocabulary

_t
_a
he
in
re

Training Data

_ P r o o f, _ o f, _t he,
_ M i l k y, _ W a y, _ c o
n s i s t i n g, _ o f, _ m
a n y, _ s t a r s, _ c a m
e, _ i n, _ 1 6 1 0, _ w he
n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _t he, _ M i l k
y, _ W a y, _a n d, _ d i s
c o v e r e d, _t h a t, _ i
t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a
i n t, _ s t a r s .

Pair counts

_t he	5605612
o n	5261131
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_ c	3831635
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Vocabulary

_t
_a
he
in
re

Training Data

_ P r o o f, _ o f, _t he,
_ M i l k y, _ W a y, _ c o
n s i s t i n g, _ o f, _ m
a n y, _ s t a r s, _ c a m
e, _ i n, _ 1 6 1 0, _ w he
n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _t he, _ M i l k
y, _ W a y, _a n d, _ d i s
c o v e r e d, _t h a t, _ i
t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a
i n t, _ s t a r s .

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_ c	3831635
n d	3811435

Vocabulary

_t
_a
he
in
re
_the

Training Data

_ P r o o f, _ o f, _the, _
M i l k y, _ W a y, _ c o n
s i s t i n g, _ o f, _ m a
n y, _ s t a r s, _ c a m
e, _ in, _ 1 6 1 0, _ w h e
n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _the, _ M i l k y,
_ W a y, _a n d, _ d i s c
o v e r e d, _t h a t, _ i
t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a
i n t, _ s t a r s .

Pair counts

_t he	5605612
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_ c	3831635
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Vocabulary

_t
_a
he
in
re
_the

Training Data

_ P r o o f, _ o f, _the, _
M i l k y, _ W a y, _ c o n
s i s t i n g, _ o f, _ m a
n y, _ s t a r s, _ c a m
e, _ in, _ 1 6 1 0, _ w h e
n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _the, _ M i l k y,
_ W a y, _a n d, _ d i s c
o v e r e d, _t h a t, _ i
t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a
i n t, _ s t a r s .

Pair counts

o n	5261131
_ o	5163783
_ s	5035505
e r	4754849
_ w	4523998
a t	4424733
o u	3838417
_ c	3831635
n d	3811435
o r	3661288

Vocabulary

_t
_a
he
in
re
_the

Training Data

_ P r o o f, _ o f, _the, _
M i l k y, _ W a y, _ c o n
s i s t i n g, _ o f, _ m a
n y, _ s t a r s, _ c a m
e, _ in, _ 1 6 1 0, _ w h e
n, _ G a l i l e o, _ G a l
i l e i, _ u s e d, _a, _t
e l e s c o p e, _t o, _ s
t u d y, _the, _ M i l k y,
_ W a y, _a n d, _ d i s c
o v e r e d, _t h a t, _ i
t, _ i s, _ c o m p o s e
d, _ o f, _a, _ h u g e, _
n u m b e r, _ o f, _ f a
i n t, _ s t a r s .

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_ c	3831635
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o r	3661288

Vocabulary

_t
_a
he
in
re
_the
⋮
*until we reach
desired vocab size T*

BPE Exercise

- Training corpus:
 - hug: 10
 - pug: 5
 - pun: 12
 - bun: 4
 - hugs: 5
- Desired vocabulary size: 10

BPE Segmentation Strategy

- Basic functioning:
 - Split each word into its characters (or bytes)
 - Follow the **merge rules** found during vocabulary creation
- Toy example: bug $\rightarrow$ b u g $\rightarrow$ b ug

Questions?